TASK 5 — PYTHON EDA CODE

Exploratory Data Analysis • Titanic Dataset

Run this code in Jupyter Notebook in the same folder as train.csv. The charts and printed summaries support the findings in the EDA report.

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# TASK 5: EXPLORATORY DATA ANALYSIS (EDA)
# Dataset: train.csv (Titanic dataset)

import pandas as pd
import matplotlib.pyplot as plt

# 1. Load and inspect the dataset
df = pd.read_csv("train.csv")
print("Shape:", df.shape)
print(df.head())
print(df.info())
print(df.describe(include="all"))
print("Missing values:\n", df.isnull().sum())
print("Duplicate rows:", df.duplicated().sum())
print("Survival counts:\n", df["Survived"].value_counts())
print("Survival rate:", round(df["Survived"].mean() * 100, 2), "%")

# 2. Visualize survival counts
df["Survived"].map({0: "Did not survive", 1: "Survived"}).value_counts().plot(
    kind="bar", title="Passenger Survival Count"
)
plt.ylabel("Number of passengers")
plt.xlabel("")
plt.tight_layout()
plt.show()

# 3. Compare survival rates by sex
(df.groupby("Sex")["Survived"].mean() * 100).plot(
    kind="bar", title="Survival Rate by Sex"
)
plt.ylabel("Survival rate (%)")
plt.tight_layout()
plt.show()

# 4. Compare survival rates by passenger class
(df.groupby("Pclass")["Survived"].mean() * 100).plot(
    kind="bar", title="Survival Rate by Passenger Class"
)
plt.ylabel("Survival rate (%)")
plt.xlabel("Passenger class")
plt.tight_layout()
plt.show()

# 5. Explore age distribution
df["Age"].dropna().plot(kind="hist", bins=20, title="Passenger Age Distribution")
plt.xlabel("Age")
plt.ylabel("Frequency")
plt.tight_layout()
plt.show()

# 6. Numeric correlation heatmap (Matplotlib only)
numeric = df[["Survived", "Pclass", "Age", "SibSp", "Parch", "Fare"]]
corr = numeric.corr()
plt.imshow(corr, aspect="auto")
plt.colorbar(label="Correlation coefficient")
plt.xticks(range(len(corr.columns)), corr.columns, rotation=35, ha="right")
plt.yticks(range(len(corr.index)), corr.index)
plt.title("Correlation Heatmap")
plt.tight_layout()
plt.show()

# 7. Grouped summaries
print("Survival rate by sex (%):")
print((df.groupby("Sex")["Survived"].mean() * 100).round(2))
print("Survival rate by passenger class (%):")
print((df.groupby("Pclass")["Survived"].mean() * 100).round(2))

# 8. Suggested cleaning notes:
# - Age has missing values; consider median imputation by relevant groups.
# - Cabin is highly incomplete; assess whether cabin deck can be extracted.
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# - Embarked has a small number of missing values; handle transparently.
# - Do not remove outliers without checking whether they are valid observations.
```